

A PLAIN-LANGUAGE STORY, START TO FINISH

What is an AI agent, really?

Follow one task, from a simple question to a finished job

Che Guan · July 18, 2026

Start with a Familiar Question

“Book me a table for four on Friday, somewhere everyone will eat.”

- A chatbot writes you a nice reply — and stops
- To actually do it, something must check the date, remember your vegetarian friend, search, decide, and book
- That gap — between answering and doing — is the whole subject of this deck

Takeaway An agent is what fills the gap between a good answer and a finished task.

Where We Are Going

The single agent

- What separates an agent from a chatbot
- Inside one agent — the parts of its “mind”
- How people actually build one

Beyond one agent

- Many agents working together
- Does it work, and is it safe?
- What is new since these surveys

Takeaway One story in five moves: what, inside, build, measure, and where it is going.

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From Chatbot to Agent

What actually makes the difference

- A chatbot answers; an agent acts
- Agent vs. workflow vs. single prompt
- Why this became practical only recently

A Chatbot Answers; an Agent Gets Things Done

Answers a question

- A chatbot
 - Reads your words, writes a reply
 - Forgets once the chat scrolls away
 - “Reading ticker tape in a dark room”

Finishes a task

- An agent
 - Perceives, remembers, plans, acts
 - Runs in a loop until the task is done
 - Can search, click, call tools, check itself

Takeaway Same language model underneath — the agent adds the parts that let it act in the world.

Agent vs. Workflow vs. Single Prompt

Single prompt

One model call, no loop

“Summarize this report” — one answer

Workflow

Fixed steps a developer
wired

*fetch → summarize → email, every
time*

Agent

The model decides the steps

chooses to search, read, then book

More autonomy → more flexible, but harder to predict and test

Takeaway Reach for an agent only when the task truly needs it.

Distinction follows Anthropic, “Building Effective Agents” (2024): workflows use fixed paths; agents direct their own.

Why This Became Practical Only Recently

Reasoning

The engine

- Models can plan and self-correct
- Not just autocomplete anymore
- Lets an agent think in steps

Tool use

The hands

- Models can call functions and APIs
- Search, run code, click buttons
- Reach beyond text into action

Memory

The thread

- Keep context across a whole task
- Recall earlier steps and facts
- Turns single replies into work

Takeaway Reasoning, tool use, and memory arrived together — that is what made agents suddenly work.

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Inside One Agent

The parts of its “mind”

- One ordinary walk home uses every part
- Perceive, think, act — on a loop
- Memory — the part that remembers you
- Plus planning, world model, motivation, tools

One Ordinary Walk Home Uses Every Part

On the way home

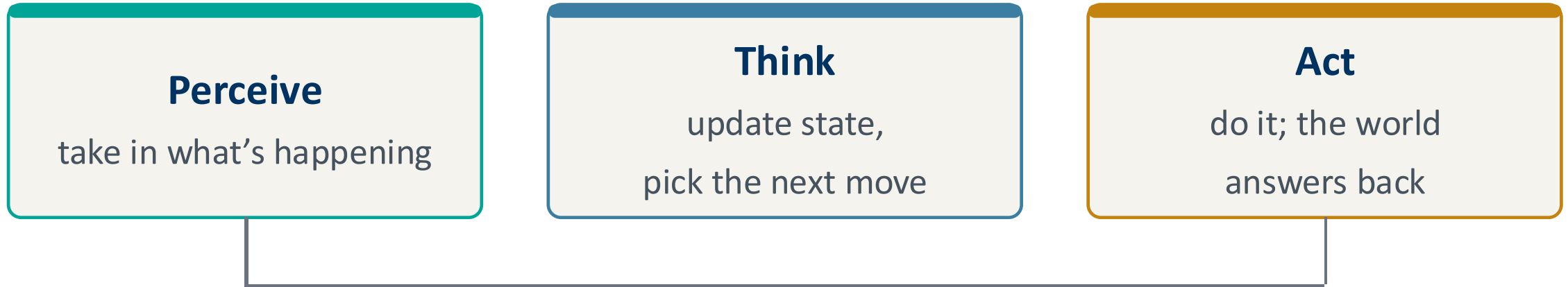
- Sense, recall, imagine
 - Eyes check traffic — perception
 - You recall a shortcut — memory
 - You picture a detour — world model

...and every trip after

- Want, decide, improve
 - Hunger pulls you to a deli — reward
 - You choose the route — planning
 - Next time is smoother — learning

Takeaway Every part of an agent is something you already do without thinking on the walk home.

Perceive, Think, Act — on a Loop



“Think” is where memory, planning, a world model, and motivation all live

Takeaway The loop is simple; the richness is all in the middle step.

Memory: the Agent That Remembers You

- Monday, you mention “I’m vegetarian” while booking lunch
- Friday, booking dinner, it skips the steakhouse on its own
- Short-term holds this task; long-term carries facts across sessions
 - Retrieval is the skill of pulling the right memory at the right moment

Takeaway Without memory every task starts from zero; with it, the agent accumulates a relationship.

Memory, Looked at More Closely

Forms

What carries it

- As text you can read
- As weights in the model
- As vectors it reads

Functions

The job it does

- Facts about you and the world
- Experience — skills and lessons
- Working notes for the task

Dynamics

How it changes

- Form new memories
- Update and merge
- Forget the stale

Takeaway Forgetting is a feature, not a bug — stale notes drown the agent.

Source: Hu et al., "Memory in the Age of AI Agents" (arXiv:2512.13564, 2026).

Before It Books, the Agent Thinks Ahead

Guess once

- The simple way: chaining
 - Pick a place, then check it fits
 - One ordered guess, straight through
 - Fast, but no second-guessing

Look ahead, then choose

- The careful way: tree search
 - Imagine a few options first
 - Weigh each, then commit to one
 - “Wait — the vegetarian friend”

Takeaway Thinking ahead is just imagining options before you commit to one.

Some models now do this visibly: DeepSeek-R1 (arXiv:2501.12948, 2025) reasons step by step and self-corrects.

A World Model Is the Mind's Eye

- A table-tennis player predicts the ball's path before the paddle moves
- An agent runs an internal “what-if” before committing to an action
- Cheap rehearsal in the head beats costly trial and error in the world
 - Risk: a bad model dreams up confident nonsense

Takeaway Imagining a step is cheaper than taking a wrong one — if the imagination is accurate.

Motivation: the Why Behind the What

What to want

- Reward — a running scoreboard
 - Each action gets a quick tip: helped or hurt
 - Gold stars from outside vs. curiosity within
 - Curiosity is the agent's own pep talk

How wanting feels

- Emotion — signal, not noise
 - Nudges priorities and how carefully it acts
 - Feeling and reasoning are intertwined
 - Poorly chosen goals invite reward hacking

Takeaway Give an agent the wrong goal and it will pursue it perfectly — the paper-clip cautionary tale.

Tools Are the Agent's Hands

- Faced with 17×23 , a good agent calls a calculator instead of guessing
- Tools = functions, search, code, other apps — the agent's reach into the world
- The frontier problem: every agent + every tool = a mess of custom glue
 - A shared standard (MCP) is “one plug for every tool”

Takeaway A model that can call tools stops guessing and starts doing.

On-slide: Model Context Protocol (Anthropic, 2024) — an open standard now adopted across the industry to connect agents to tools.

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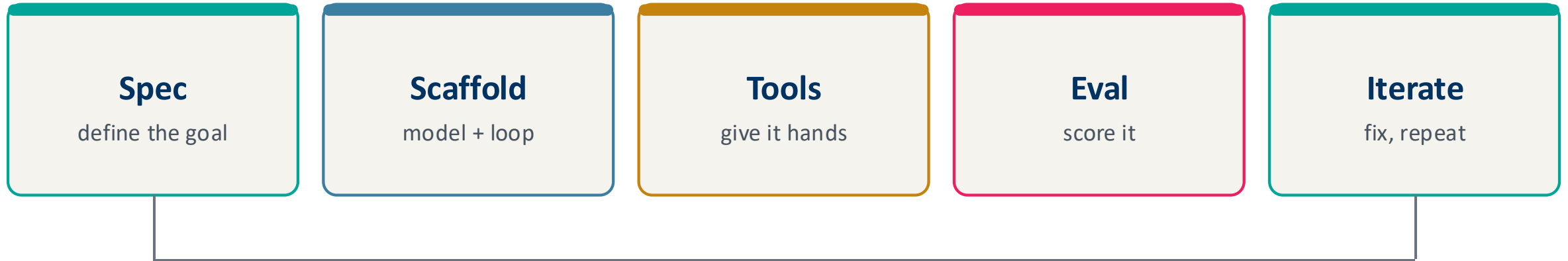
PART

From One Agent to Many

Building, scaling, improving

- How you actually build one
- Many agents, three ways to organize
- Agents that improve themselves

How You Actually Build One



Evaluation feeds back into the spec — building an agent is a loop, not a straight line

Takeaway Start simple, add autonomy only where evaluation shows it earns its cost.

Many Agents, Three Ways to Organize

Centralized

A manager and workers

- One agent plans and assigns
- Easy to steer and follow
- Single point of failure

Decentralized

Peers talking

- Agents message and critique
- Robust, scales out
- Harder to steer and debug

Hybrid

A bit of both

- Central rules, local teams
- Balance control + flexibility
- e.g. CAMEL, DyLAN

Takeaway Three simple shapes — and one warning: a single weak agent can sink the team.

Agents That Improve Themselves

The mechanisms

- How agents get better
 - Reflect on failures and correct
 - Bank reusable skills for later
 - Learn from feedback and rivals

The mechanisms in action

- Vivid case: Voyager in Minecraft
 - Plays, then writes skills to a library
 - Reuses them to tackle harder quests
 - Competence compounds with experience

Takeaway The best agents do not just run — they get better at running, one task to the next.

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Does It Work, and Is It Safe?

Measuring and securing agents

- Scoring an agent is harder than scoring a model
- Attacks on the brain, and through the world
- Where agents already earn their keep

Scoring an Agent Is Harder than Scoring a Model

- A model gives one answer; an agent takes a whole path of decisions
- Ask two questions: did it succeed, and was the path sound?
- Environments are noisy — the same task can pass once and fail next run
 - No single score captures it — hence a crowded benchmark landscape

Takeaway Grade the destination and the journey — a right answer by luck is not a good agent.

Attacks on the Brain, and Through the World

Intrinsic threats

- From inside the model
 - Jailbreaks slip past safety training
 - Prompt injection hijacks the plan
 - Poisoning corrupts the model itself

Extrinsic threats

- From its surroundings
 - A booby-trapped document it retrieves
 - Spoofed sensors — fake GPS, fake readings
 - One breach can spread across agents

Takeaway Safety can't be a filter bolted on the end — it has to be built into the goal itself.

Where Agents Already Earn Their Keep

Science

A tireless lab partner

- Plan and run experiments
- ChemCrow, GeneAgent
- Reason over domain tools

Software

A small dev team

- Write, test, review code
- ChatDev, MetaGPT
- Split roles across agents

Everyday work

A capable assistant

- Research, book, summarize
- Recommend and plan
- The dinner booking, finished

Takeaway The pattern is always a reasoner orchestrating tools — not a lone model acting by itself.

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Where Agents Are Headed

What changed since these surveys

- Reasoning you can watch happen
- A universal plug for tools
- Agents that work across many sessions
- The whole story, on one slide

What's New Since These Surveys

Reasoning models

Think, visibly

- Trained to reason by RL
- Stop and reconsider mid-answer
- DeepSeek-R1 (2501.12948)

Tool standard

One plug for all

- MCP connects agents to tools
- Adopted across the industry
- Anthropic (2024)

Long-running agents

Work in shifts

- Span many context windows
- Hand off notes between shifts
- Anthropic harness (2025–26)

Takeaway The surveys drew the map; these are the roads built since.

DeepSeek-R1 (arXiv:2501.12948); Model Context Protocol (Anthropic, 2024); Anthropic harness-design articles (2025–2026).

A Closer Look: Agents That Work in Shifts

- A hard task can outlast the model's memory window
- Picture engineers in shifts — each arrives remembering nothing of the last
- The fix: leave clear notes so the next shift picks up cleanly
 - Sharp caveat: every harness assumes the model can't do something — and that expires

Takeaway When tasks outgrow one session, the hard part becomes the hand-off.

Anthropic, "Effective harnesses for long-running agents" (2025) and "Harness design for long-running apps" (2026).

The Whole Story, on One Slide

One agent

The single mind

- Chatbot answers; agent acts
- Memory, planning, world model
- Motivation and tools

Built and scaled

From one to many

- Spec → scaffold → eval loop
- Three ways to organize teams
- Agents that improve themselves

Measured and safe

Kept honest

- Grade path and outcome
- Threats inside and out
- Safety built into the goal

Takeaway Remember the walk home: everything an agent does, you already do without thinking.

Sources and Further Reading

- Luo et al. (2025), LLM Agent survey — engineering view · arXiv:2503.21460
- Liu et al. (2025), Foundation Agents — brain-inspired view · arXiv:2504.01990
- Hu et al. (2026), Memory in the Age of AI Agents · arXiv:2512.13564
 - Since the surveys: DeepSeek-R1 (2501.12948); MCP (Anthropic, 2024); Anthropic harness-design articles (2025–26)
 - Every named system and claim is the source authors' own, not independently verified here